Anonymous and Neutral Classification Aggregation*

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In classification aggregation, individuals classify objects into categories and the aim is to reach an aggregate classification. In preference aggregation, individuals rank alternatives and the aim is to reach an aggregate ranking. In this paper, we highlight the inherent connection between these two problems. In particular, we show that what we know about the difficulty of having anonymous and neutral preference aggregation can be imported into classification aggregation.

1 Introduction

Preference aggregation is a well-known and broadly studied problem where individual preferences on a set of alternatives are aggregated into a societal preference. Classical works in this strand of literature start with the impossibility theorem by Arrow [1951] that shows that independence and Pareto

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dominance are not compatible with nondictatorial aggregation. Consequent works extensively studied related axiomatics, such as, among others, Wilson [1972], who demonstrates a similar impossibility result under independence and citizen sovereignty, and Moulin [1980], under anonymity and neutrality.

The aforementioned results are in the context of aggregating preferences expressed as rankings of a set of objects. The classification aggregation problem, on the other hand, deals with aggregating individual classifications of objects into a societal classification. A classification is an assignment of a set of objects into a set of categories such that no category is left unassigned. Rather surprisingly, scholarly interest in this problem has only emerged very recently. Namely, Maniquet and Mongin [2016] proved an *Arrovian* impossibility theorem (henceforth, *MM result*) in the context of classification aggregation that establishes that independent and unanimous aggregation is necessarily dictatorial. Building on this, Cailloux et al. [2023] recently delivered a *Wilsonian* impossibility result in this setting, namely, one that replaces unanimity with citizen sovereignty. Furthermore, Maniquet and Mongin [2014] argue that the distinguished capacity of judgment aggregation approach to capture nonbinary settings could be employed to obtain results in novel contexts such as classification aggregation, *e.g.,* the MM result [see Grossi and Pigozzi, 2022, for a primer in judgment aggregation theory].

We concur that there remains much to learn about classification aggregation through the lenses of established approaches in preference or judgment aggregation, as evidenced by our recent work [Cailloux et al., 2023]. In this paper, we study what the axiomatics of Moulin [1980] imply in the context of classification aggregation when the number of objects equals that of categories. Specifically, we prove that the Moulin impossibility result entails an equivalent result in classification aggregation, and furthermore propose a greedy aggregation rule based on Doğan and Giritligil [2022] for the instances when anonymity and neutrality are compatible.

In our analysis we highlight the similarity between classification aggregation and preference aggregation. In particular, we prove that the problem of preference aggregation with an imposed number of equivalence classes is isomorphic to classification aggregation. Imposing a maximal number of equivalence classes for preference aggregation has been broadly studied in the literature with the examples of approval voting [Brams and Fishburn, 1978], range voting [Smith, 2000], or majority judgement [Balinski and Laraki, 2011]. The only work we know that imposes a fixed number of equivalence classes is by Maniquet and Mongin [2015], which focuses on approval voting (2 equivalence classes). In practical applications, a very popular example is that of aggregating tier-lists in gaming.[1] Some tiers are predefined and each gamer defines a mapping from the set of candidates to the set of tiers, establishing a tier-list. A tier-list that leaves a tier empty is usually considered inconsistent, so the number of tiers (categories) is, in general, fixed.

To motivate our formal analysis that follows, imagine a theater club that consists of n teachers and m actors (students). Not to leave out any actor, the teachers have decided on a play that has m roles, and they need to assign roles to actors. Each teacher has an idea on how to attribute these roles. The procedure to reach an aggregate assignment has to be fair vis-a-vis the teachers and the actors; hence no differentiation among teachers and actors. This problem can be thought of as a classification aggregation problem with equal number of objects and categories, where anonymity and object neutrality are desired: identities of actors or teachers should not have any impact on the final attribution of roles. We obtain in what follows a *Moulinian* impossibility theorem in this context.

2 Preliminaries

2.1 Classification aggregation

We consider a set $N = \{1, \ldots, n\}$ of *individuals* with $n \geq 2$, a set $P = \{p_1, \ldots, p_\rho\}$ of *categories* with $\rho \geq 2$, and a set $X = \{x_1, \ldots, x_m\}$ of *objects* with $m \geq \rho$.

We define a *classification* as a surjective mapping $c : X \to P$ and denote by $\mathcal{C} \subset P^X$ the set of classifications. We write $\boldsymbol{c} = (c_1, \ldots, c_n) \in \mathcal{C}^N$ for a classification profile. Given $\boldsymbol{c} \in \mathcal{C}^N$ and $x \in X$, we write $\boldsymbol{c}_x \in P^N$ for the vector of categories that object x is put into by each individual, thus $\forall i \in N, \boldsymbol{c}_x(i) = c_i(x)$. A *classification aggregation function* (CAF) is a mapping $\alpha : \mathcal{C}^N \to \mathcal{C}$.

We let S_X, S_N denote the sets of permutations of respectively X and N. Given a classification profile $\boldsymbol{c} \in \mathcal{C}^N$ and a permutation $\gamma \in S_N$, we introduce $\boldsymbol{c}^{(\gamma)} = (c_{\gamma(1)}, \ldots, c_{\gamma(n)})$. Note that a classification profile $\boldsymbol{c}$ can be seen as a function from N to $\mathcal{C}$ or from X to P^N or from $N \times X$ to P. Using this fact, given $\pi : P \leftrightarrow P$ we will write $\pi \circ \boldsymbol{c}$ to denote the classification profile equal to $(\pi \circ c_i)_{i \in N}$.

A CAF α is said to be *anonymous* if $\forall \boldsymbol{c} = (c_1, \ldots c_n) \in \mathcal{C}^N, \forall \pi : N \leftrightarrow N, \alpha(c_{\pi(1)}, \ldots, c_{\pi(n)}) = \alpha(\boldsymbol{c})$. A CAF α is *object neutral* if $\forall x, y \in X, \boldsymbol{c}, \boldsymbol{c}' \in \mathcal{C}^N, (\boldsymbol{c}_x = \boldsymbol{c}'_y) \wedge (\boldsymbol{c}_y = \boldsymbol{c}'_x) \wedge (\forall z \in X \setminus \{x, y\}, \boldsymbol{c}_z = \boldsymbol{c}'_z) \implies \alpha(\boldsymbol{c})(x) = \alpha(\boldsymbol{c}')(y)$.

[1] `https://tiermaker.com`

Note that as any permutation can be decomposed into swaps, object neutrality implies

$$\forall \pi : X \leftrightarrow X, \boldsymbol{c} \in \mathcal{C}^N, \alpha((\boldsymbol{c}_{\pi(x)})_{x \in X}) = \alpha((\boldsymbol{c}_x)_{x \in X}) \circ \pi = \alpha(\boldsymbol{c}) \circ \pi.$$

A CAF α is *category neutral* if $\forall \pi : P \leftrightarrow P, \forall \boldsymbol{c} \in \mathcal{C}^N, \alpha(\pi \circ \boldsymbol{c}) = \pi \circ \alpha(\boldsymbol{c})$.

2.2 Preference aggregation

We also consider a set N of voters or individuals and a set X of objects, with $m = |X| \geq 2$. We let $\mathcal{L}(X)$ the set of linear orders, i.e., complete, transitive and asymmetric binary relations on X. Given a voter $i \in N$, we write $P_i \in \mathcal{L}(X)$ for the preference of voter i on X. A n-tuple P_N in $\mathcal{L}(X)^N$ is called a preference profile.

A social welfare function (SWF) is a mapping from $\mathcal{L}(X)^N$ to $\mathcal{L}(X)$. Given a preference profile $P_N \in \mathcal{L}(X)^N$ and a permutation $\gamma \in S_N$, we write $\gamma(P_N) = (P_{\gamma(i)})_{i \in N}$ for the profile induced by the permutation of voters. A SWF f is said to satisfy *welfare anonymity* if $\forall \gamma \in S_N, \forall P_N \in \mathcal{L}(X)^N, f(\gamma(P_N)) = f(P_N)$.

By abuse of notation, given an individual preference $P_i \in \mathcal{L}(X)$ and a permutation $\pi \in S_X$, let $\pi(P_i)$ be such that $x \ \pi(P_i) \ y \iff x \ P_i \ y$ for all $x, y \in A$. Given a preference profile $P_N \in \mathcal{L}(X)^N$ and a permutation $\pi \in S_X$, we write $\pi(P_N) = (\pi(P_i))_{i \in N}$ for the profile induced by a permutation of objects π in P_N. A SWF f is said to satisfy *welfare neutrality* if $\forall \pi \in S_X, \forall P_N \in \mathcal{L}(X)^N, f(\pi(P_N)) = \pi(f(P_N))$.

We introduce $\mathcal{W}_\rho(X)$ to be the set of weak orders of X with exactly ρ equivalence classes. A ρ-SWF is a mapping from $\mathcal{W}_\rho(X)^N$ to $\mathcal{W}_\rho(X)$.

2.3 Isomorphism

We claim that $\mathcal{C}$ and $\mathcal{W}_\rho(X)$ are isomorphic. Indeed, given $P = \{p_1, \ldots, p_m\}$ and an arbitrary strict ordering $\succ$ of P we can define $\theta_\succ : \mathcal{C} \to \mathcal{W}_\rho(X)$ as $\forall c \in \mathcal{C}, \forall x, y \in X, x \ \theta_\succ(c) \ y$ *iff* $c(x) \succ c(y)$. One can check that $\theta_\succ$ is an isomorphism. Note that in the case of $m = \rho$, we have $\mathcal{L}(X) = \mathcal{W}_\rho(X)$, so that $\mathcal{C}$ and $\mathcal{L}(X)$ are isomorphic. In what follows, we will suppose that this condition holds. In what follows, we let θ be any isomorphism from $\mathcal{C}$ to $\mathcal{L}(X)$ and φ an isomorphism from $\mathcal{L}(X)$ to $\mathcal{C}$.

Remark 1. A classification c can be viewed as a function that sends each object x to a number (namely the number k corresponding to $p_k = c(x)$). Similarly, a weak order P_i having ρ equivalence classes can be viewed as a function that sends each object x to a number in $[\![1, \rho]\!]$ (namely the rank of x in P_i, corresponding to the cardinality of the number of equivalence classes in its weak upper contour set, weak meaning including itself). By doing so, we can build an isomorphism from $\mathcal{C}$ to $\mathcal{W}_\rho(X)$ by using the number that objects are mapped to. $\triangle$

Note that the isomorphism θ between $\mathcal{C}$ and $\mathcal{L}(X)$ induces an isomorphism between $\mathcal{C}^N$ and $\mathcal{L}(X)^N$ and therefore between the set of CAFs and the set of SWFs. By abuse of notation, given a CAF α, we let $\theta(\alpha)$ denote the SWF defined as $\forall \boldsymbol{c} = (c_i)_{i\in N} \in \mathcal{C}^N, \theta(\alpha)((\theta(c_i))_{i\in N}) = \theta(\alpha(\boldsymbol{c}))$. Likewise, given an SWF f, we define $\varphi(f)$ as $\forall P_N \in \mathcal{L}(X), \varphi(f)((\varphi(P_i))_{i\in N}) = \varphi(f(P_N))$.

Note that given a permutation $\pi \in S_X$, θ and π are associative, i.e. $\forall c \in \mathcal{C}, \theta(c \circ \pi) = \pi(\theta(c))$.

3 Results

Proposition 1. *A CAF α satisfies anonymity iff $\theta(\alpha)$ satisfies welfare anonymity.*

Proof. Let α be a CAF.

$\Rightarrow$ Suppose α satisfies anonymity, and let $\gamma \in S_N$ be a permutation of N. We have $\forall \boldsymbol{c} = (c_i)_{i\in N} \in \mathcal{C}^N, \alpha((c_{\gamma(i)})_{i\in N}) = \alpha(\boldsymbol{c})$. Let $P_N \in \mathcal{L}(X)$ be any preference profile, then $\exists \boldsymbol{c} \in \mathcal{C}, \theta(\boldsymbol{c}) = P_N$ as θ is an isomorphism. Therefore, $\theta(\alpha)(\gamma(P_N)) = \theta(\alpha)((\theta(c_{\gamma(i)}))_{i\in N}) = \theta(\alpha(\boldsymbol{c}^{(\gamma)})) = \theta(\alpha(\boldsymbol{c})) = \theta(\alpha)((\theta(c_i))_{\in N}) = \theta(\alpha)(P_N)$ as α satisfies anonymity, so $\alpha(\boldsymbol{c}^{(\gamma)}) = \alpha(\boldsymbol{c})$. Then, $\theta(\alpha)$ satisfies welfare anonymity.

$\Leftarrow$ Suppose $\theta(\alpha)$ satisfies welfare anonymity, and let $\boldsymbol{c} = (c_i)_{i\in N} \in \mathcal{C}^N$ be a classification profile and $\gamma \in S_N$ be a permutation. As $\theta(\alpha)$ satisfies welfare anonymity, we have $\theta(\alpha)(\gamma((\theta(c_i))_{i\in N})) = \theta(\alpha)((\theta(c_i))_{i\in N})$.

Then $\theta(\alpha(\boldsymbol{c}^{(\gamma)})) = \theta(\alpha)(\gamma((\theta(c_i))_{i\in N})) = \theta(\alpha)((\theta(c_i))_{i\in N}) = \theta(\alpha(\boldsymbol{c}))$. As θ is an isomorphism, $\theta(\alpha(\boldsymbol{c}^{(\gamma)})) = \theta(\alpha(\boldsymbol{c}))$ implies $\alpha(\boldsymbol{c}^{(\gamma)}) = \alpha(\boldsymbol{c})$, so α satisfies anonymity. $\square$

Corollary 1. *A SWF f satisfies welfare anonymity iff $\varphi(f)$ satisfies anonymity.*

Proposition 2. *A CAF α satisfies object neutrality iff $\theta(\alpha)$ satisfies welfare neutrality.*

Proof. $\Rightarrow$ Suppose α satisfies object neutrality, and let $\pi \in S_X$ be a permutation of X.

We have that $\forall \boldsymbol{c} \in \mathcal{C}^N, \alpha((c_i \circ \pi)_{i\in N}) = \alpha(\boldsymbol{c}) \circ \pi$. Now take $P_N = (P_i)_{i\in N} \in \mathcal{L}(X)^N$, as θ is an isomorphism, $\exists \boldsymbol{c} = (c_i)_{i\in N} \in \mathcal{C}^N, \forall i \in N, \theta(c_i) = P_i$.

Now, $\theta(\alpha)(\pi(P_N)) = \theta(\alpha)(\pi((\theta(c_i)))_{i\in N}) = \theta(\alpha)(\theta((c_i \circ \pi))_{i\in N}) = \theta(\alpha((c_i \circ \pi)_{i\in N})) = \theta(\alpha(\boldsymbol{c}) \circ \pi) = \pi(\theta(\alpha)(P_N))$. Then, $\theta(\alpha)$ satisfies welfare neutrality.

$\Leftarrow$ Suppose $\theta(\alpha)$ satisfies welfare neutrality, and let π be a permutation of X.

We have that $\forall P_N \in \mathcal{L}(X)^N, \theta(\alpha)(\pi(P_N)) = \pi(\theta(\alpha)(P_N))$. Now, let $\boldsymbol{c} = (c_i)_{i\in N} \in \mathcal{C}^N$, and consider $P_N = (\theta(c_i))_{i\in N} \in \mathcal{L}(X)^N$.

We have $\theta(\alpha((c_i \circ \pi)_{i\in N})) = \theta(\alpha)((\theta(c_i \circ \pi)_{i\in N}) = \theta(\alpha)((\pi(\theta(c_i)))_{i\in N}) = \theta(\alpha)(\pi(P_N)) = \pi(\theta(\alpha)(P_N)) = \theta(\pi(\alpha(\boldsymbol{c})))$. Then, as θ is an isomorphism, $\alpha((c_i \circ \pi)_{i\in N}) = \alpha(\boldsymbol{c}) \circ \pi$, so α satisfies object neutrality. $\square$

Corollary 2. *A SWF f satisfies welfare neutrality iff $\varphi(f)$ satisfies object neutrality.*

Theorem 1 (Moulin [1980]). *Given $n, m \geq 2$, there exists a welfare anonymous and welfare neutral SWF iff all prime divisors of n exceed m.*

Corollary 3. *Given $n, m = \rho \geq 2$, there exists an anonymous and object neutral CAF iff all prime divisors of n exceed m.*

Doğan and Giritligil [2022] defined a SWF in their theorem 3.1 that they proved to be well defined and to satisfy welfare anonymity and welfare neutrality whenever all prime divisors of n exceed m. We refer to it as the greedy SWF.

Algorithm 1 The greedy SWF

while $\exists P_N \in \mathcal{L}(X)^N, f(P_N)$ is not defined **do**
  Pick $P_N \in \mathcal{L}(X)^N, f(P_N)$ is not defined, and $l \in \mathcal{L}(X)$
  Set $f(P_N) = l$
  for $\tilde{P_N} \in \mathcal{L}(X)^N$ s.t. $\exists \pi \in S_X, \gamma \in S_N, \forall i \in N, \tilde{P}_i = \pi(P_{\gamma(i)})$ **do**
    $f(\tilde{P_N}) = \pi(l)$
  end for
end while

We now introduce the greedy anonymous and neutral CAF which is isomorphic to the greedy SWF:

Algorithm 2 The greedy CAF

while $\exists \boldsymbol{c} \in \mathcal{C}^N, \alpha(\boldsymbol{c})$ is not defined **do**
  Pick $\boldsymbol{c} \in \mathcal{C}^N, \alpha(\boldsymbol{c})$ is not defined, and $\tilde{c} \in \mathcal{C}$
  Set $\alpha(\boldsymbol{c}) = \tilde{c}$
  for $\boldsymbol{c}' \in \mathcal{C}^N$ s.t. $\exists \pi \in S_X, \gamma \in S_N, \forall i \in N, c'_i = c_{\gamma(i)} \circ \pi$ **do**
    $\alpha(\boldsymbol{c}') = \alpha(\boldsymbol{c}) \circ \pi$
  end for
end while

Proposition 3. *The greedy anonymous and neutral CAF satisfies anonymity and neutrality iff all prime divisors of n exceed m.*

4 Conclusion

For the problem of classification aggregation where the number of objects matches the number of categories, we were able to prove an equivalent of Moulin's [1980] impossibility result in classification aggregation through an isomorphism between the set of CAFs and the set of SWFs. In order to complete the picture, one should look at what happens when the number of objects exceeds the number of categories. The existence of anonymous and neutral CAFs in this case remains an open question.

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